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Variation in Form of the Pyramidal Tract and Its Relationship to Digital Dexterity[1]

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Key Words. Pyramidal tract · Corticospinal tract · Digital dexterity · Mammals · Primates

Abstract. A morphometric analysis of the pyramidal tract's relation to digital dexterity was performed on data from 69 mammals. The results show that the variation in digital dexterity among mammals corresponds most closely to the variation in place of termination of pyramidal tract fibers within the spinal cord, corresponds less closely to the variation in the size of the tract itself and its constituent fibers, and does not correspond reliably with any other feature yet reported. Since the termination of pyramidal tract fibers on or very near spinal motor neurons is a prerequisite even for the peculiar kind of dexterity seen in some non-primates (e.g., raccoon, kinkajou), this one feature alone seems to be a critical factor.

Introduction

On the basis of ablation studies on scattered mammalian species, there is a puzzling dichotomy in the function of the pyramidal tract between anthropoids and other mammals. In man or monkey, interruption of the pyramidal tract results in a weakness graded in increasing severity from proximal to distal and a loss of the ability to move the digits independently [TOWER, 1940; BUCY *et al.*, 1966; LAWRENCE and KUYPERS, 1968a, b]. In marked contrast, direct section of the pyramidal tract or ablation of its cortical origin in non-anthropoids has little apparent effect. In dogs, cats, prosimians, phalangers, or opossums, pyramidal tract

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destruction results in an initial weakness sometimes lasting several weeks, but it does not result in a paralysis reminiscent of that seen in anthropoids nor in any obvious or permanent deficit [BROMILEY and BROOKS, 1940; BUXTON and GOODMAN, 1967; CHAMBERS and LIU, 1957; HORE *et al.*, 1973; JANE *et al.*, 1965; REES and HORE, 1970].

The increase in anatomical information over the last 20 years [LASSEK, 1954] makes it possible to suggest that the marked differences in type or degree of deficit parallel the fine gradations in pyramidal tract anatomy. In general, the mammals which suffer the greatest deficit as a result of damage to their pyramidal tract are those with the most highly developed digital skills; and those which are most dexterous seem to be those with the most highly developed pyramidal tracts. It is this latter relationship between the variation in form of the pyramidal tract and the variation in degree of digital dexterity which is the primary focus of this report.

During the past half century, numerical data describing the pyramidal tract in a variety of mammals have become available. This pool of data has now grown to a point which allows a strict comparative inquiry using only commonplace statistical techniques. Accordingly, we have collected an array of from two to nine sets of descriptive data on pyramidal tract morphology and digital dexterity in a total of 69 mammalian species and subjected it to a statistical analysis which serves to identify several reliable relationships between digital dexterity and variations in the pyramidal tract or its sites of termination.

Materials and Methods

The data for this report were derived almost entirely from the published work of others. The reports contributing numerical data are listed in table I and designated in the bibliography by an asterisk. In general, three classes of descriptive parameters were specifically included in the analysis: (1) two parameters describing the animals themselves (body weight and body length); (2) four parameters describing the pyramidal tract and its constituent fibers (area of the tract, number of fibers per tract, largest fiber diameter, and average fiber size), and (3) three parameters describing the tract's terminations in the spinal cord (depth of penetration down the spinal cord, deepest, or ventralmost, lamina in the spinal gray with pyramidal tract terminations, and lamina in the spinal gray receiving the densest pyramidal tract terminations). The relationships between these parameters were examined first singly in both linear and logarithmic forms, and then in several combinations, for their power to account for digital dexterity.

Table I. Animals and anatomical data on which all correlations are based

Animal	Digital dexterity	Body weight kg	Number of fibers per tract	Largest fiber diameter μm	Average fiber size $\times 10^5$ K	Area of tract mm^2	Penetration down spinal cord	Lamina of axon terminals: deepest	Lamina of axon terminals: densest
Aardvark *Orycteropus afer*	2	60				0.53 VERHAART, 1970	Upper C VERHAART, 1970		
Anteater *Myrmecophaga jubata*	1	20		10 VERHAART, 1970		0.60 VERHAART, 1970			
Armadillo *Dasypus novemcinctus*	1	5		2 VERHAART, 1970		0.60 VERHAART, 1970	T3 FISHER *et al.*, 1969	8 FISHER *et al.*, 1969	6
Bat (vampire) *Desmodus rufus*	1	0.033				0.01 YAMAMOTO *et al.*, 1955			
Bat (Indian tuber nose) *Harpyiocephalus leucogostra*	1	0.01				0.04 YAMAMOTO *et al.*, 1955			
Bat *Myotis*	1	0.03				0.01 VERHAART, 1970	S BROERE, 1966		
Bat (India Kalong) *Pteropus edulis*	1	0.9				0.12 VERHAART, 1970			

Table I (continued)

Animal	Digital dexterity	Body weight kg	Number of fibers per tract	Largest fiber diameter μm	Average fiber size $\times 10^5$ K	Area of tract mm^2	Penetration down spinal cord	Lamina of axon terminals	
								deepest	densest
Bat (horseshoe) *Rhinolophus*	1	0.016				0.01 Yamamoto *et al.*, 1955			
Bat *Tasarida cynocephala*	1	1.3	8,000		0.25	0.02 Lassek and Shapiro, 1956			
Bear (assume Brown)	2	200				9.21 Verhaart, 1970			
Beaver *Castor canadensis*	3	13					S Douglas and Barr, 1950		
Bushbaby *Galago crassicaudatus*	5	1.14					S Ferrer, 1971	9 Ferrer, 1971	6 Ferrer, 1971
Bushbaby *Galago senegalensis*	5	0.26		2.5 Verhaart, 1966			Lower L Verhaart, 1966		
Camel *Camelus*	1	570		6 Verhaart, 1970					

Table I (continued)

Animal	Digital dexterity	Body weight kg	Number of fibers per tract	Largest fiber diameter μm	Average fiber size $\times 10^5$ K	Area of tract mm^2	Penetration down spinal cord	Lamina of axon terminals: deepest	Lamina of axon terminals: densest
Cat (domestic) *Felis catus*	2	3.5	186,000 LASSEK and RASMUSSEN, 1940	5 SCHEIBEL and SCHEIBEL, 1966	0.60	1.13 LASSEK and RASMUSSEN, 1940	S NYBERG-HANSEN and BRODAL, 1963	8 NYBERG-HANSEN and BRODAL, 1963	6 NYBERG-HANSEN and BRODAL, 1963
Chimpanzee *Pan troglodytes*	6	44	807,000 LASSEK and WHEATLEY, 1945		0.96	7.77 LASSEK and WHEATLEY, 1945	coccygeal PETRAS, 1968	10 PETRAS, 1968	6 PETRAS, 1968
Chipmunk *Tamias striatus*	3	0.1				0.26 SIMPSON, 1914	S SIMPSON, 1914		
Cow (unspecified)	1	500	540,000 LASSEK, 1942	5 VERHAART, 1970	0.55	2.97 LASSEK, 1942	Upper C LASSEK, 1942		
Coypu rat *Myocastor coypus*	3	5		8 GOLDBY and KACKER, 1963		0.37 GOLDBY and KACKER, 1963	S GOLDBY and KACKER, 1963	6 GOLDBY and KACKER, 1963	
Deer (unspecified)	1	200	490,000 LASSEK, 1942	6 VERHAART, 1970	0.58	2.89 LASSEK, 1942	Upper C LASSEK, 1942		

Table I (continued)

Animal	Digital dexterity	Body weight kg	Number of fibers per tract	Largest fiber diameter μm	Average fiber size $\times 10^5$ K	Area of tract mm^2	Penetration down spinal cord	Lamina of axon terminals	
								deepest	densest
Dog (domestic) *Canis familiaris*	2	11	285,500 LASSEK and RASMUSSEN, 1940		1.22	3.49 LASSEK and RASMUSSEN, 1940	L5 BUXTON and GOODMAN, 1967	8 BUXTON and GOODMAN, 1967	
Elephant (Indian) *Elephas maximus*	1	6000		8 VERHAART, 1963		5.13 VERHAART, 1970	Middle T VERHAART, 1963		
Ferret *Putorius furo*	3	0.9	90,000 NOORDUYN, 1959	10 NOORDUYN, 1959		1.5 NOORDUYN, 1959			
Gibbon *Hylobates lar*	6	6	315,000 VERHAART, 1948b	12 VERHAART, 1970	1.09	3.44 VERHAART, 1970	S PETRAS, 1968	10 PETRAS, 1969	6 PETRAS, 1968
Gibbon *Hylobates syndactylus*	6	10	239,700 VERHAART, 1948b		1.35	3.25 VERHAART, 1948b			
Goat (domestic) *Capra hircus*	1	80	260,000 LASSEK, 1942	5 VERHAART, 1970	0.21	0.569 VERHAART, 1970	C7 HAARTSEN and VERHAART, 1967	6 HAARTSEN and VERHAART, 1967	

Table I (continued)

Animal	Digital dexterity	Body weight kg	Number of fibers per tract	Largest fiber diameter μm	Average fiber size $\times 10^5$ K	Area of tract mm^2	Penetration down spinal cord	Lamina of axon terminals	
								deepest	densest
Gopher *Spermophilus tridecemlineatus*	3	0.65					L3 SIMPSON, 1915		
Guinea Pig *Cavia aperea*	2	0.6				0.33 REVELEY, 1915			
Hamster *Cricetus auratus*	3	0.5	480,000 DOUGLAS and BARR, 1950				S DOUGLAS and BARR, 1950		
Hedgehog *Erinaceous*	2	0.75				0.02 VERHAART, 1970	Upper C VERHAART, 1970		
Hedgehog *Paraechinus*	2	0.5					Upper C ROBARDS, personal commun.; CAMPBELL, 1967		
Hog (domestic) *Sus scrofa*	1	90	210,000 LASSEK, 1942	5 VERHAART, 1970	0.85	1.8 LASSEK, 1942			
Horse (domestic) *Equus caballus*	1	350		5 VERHAART, 1970		2.68 VERHAART, 1970			

Table I (continued)

Animal	Digital dexterity	Body weight kg	Number of fibers per tract	Largest fiber diameter μm	Average fiber size $\times 10^5$ K	Area of tract mm²	Penetration down spinal cord	Lamina of axon terminals: deepest	Lamina of axon terminals: densest
Kinkajou *Potos flavus*	3	2.1					Lower L HARTING and NOBACK, 1970	10 HARTING and NOBACK, 1970	
Lemur *Lemur catta*	5	2.1		3 VERHAART, 1966			Lower L VERHAART, 1966		
Man *Homo sapiens*	7	70	1,101,000 LASSEK and RASMUSSEN, 1940	22 TRUEX and CARPENTER, 1969	1.03	11.43 LASSEK and RASMUSSEN, 1940	coccygeal TRUEX and CARPENTER, 1969		
Mole *Scalopus aquaticus*	1	0.1		4 BISHOP *et al.*, 1971		0.01 LINOWIECKI, 1914	Upper T LINOWIECKI, 1914		
Monkey (spider) *Ateleus ater*	4	6	505,000 LASSEK, 1943	12 LASSEK, 1943	0.63	3.23 LASSEK, 1943	coccygeal PETRAS, 1968	10 PETRAS, 1969	6 PETRAS, 1968
Monkey (capuchin) *Cebus apella*	5	1.25				2.0 MANOCHA *et al.*, 1968	coccygeal PETRAS, 1968	10 PETRAS, 1969	6 PETRAS, 1968
Monkey (vervet) *Cercopithecus pygerythrus*	6	4					coccygeal PETRAS, 1968	10 PETRAS, 1969	6 PETRAS, 1968

Table I (continued)

Animal	Digital dexterity	Body weight kg	Number of fibers per tract	Largest fiber diameter μm	Average fiber size $\times 10^5$ K	Area of tract mm^2	Penetration down spinal cord	Lamina of axon terminals	
								deepest	densest
Monkey (wooly) *Lagothrix lagotricha*	5	6					coccygeal PETRAS, 1968	10 PETRAS, 1968	6 PETRAS, 1968
Monkey (macaque) *Macaca ira*	6	5	195,000 VERHAART, 1948a		0.60	1.18 VERHAART, 1954			
Monkey (macaque) *Macaca mulatta*	6	8.3	400,000 DEMEYER and RUSSEL, 1958		0.72	2.89 LASSEK, 1941	coccygeal PETRAS, 1968	10 PETRAS, 1969	6 PETRAS, 1968
Monkey (baboon) *Papio papio*	6	25						10 PHILLIPS, 1967	
Monkey (marmoset) *Saquinas oedipus*	4	0.4					S SHRIVER and MATZKE, 1965	6 SHRIVER and MATZKE, 1965	
Monkey (squirrel) *Saimiri sciureus*	5	0.75		8 VERHAART, 1970		0.70 VERHAART, 1970	S VERHAART, 1966	9 HARTING and NOBACK, 1970	6 HARTING and NOBACK, 1970

Table I (continued)

Animal	Digital dexterity	Body weight kg	Number of fibers per tract	Largest fiber diameter μm	Average fiber size $\times 10^5$ K	Area of tract mm^2	Penetration down spinal cord	Lamina of axon terminals	
								deepest	densest
Mouse (unspecified)	3	0.025	32,300 Lassek and Rasmussen, 1940		0.21	0.07 Lassek and Rasmussen, 1940			
Mule *Equus*	1	250	412,000 Lassek, 1942	5 Verhaart, 1970	0.90	3.74 Lassek, 1942			
Opossum *Didelphis*	3	3.7	48,100 Towe, 1973		0.45	0.22 Towe, 1973	Upper T Bautista and Matzke, 1965	8 Martin and Fisher, 1968	5 Martin and Fisher, 1968
Orangutan *Pongo pygmaeus*	6	60				2.61 Verhaart, 1970			
Pangolin *Manis pentadactyla*	1	18		2.5 Verhaart, 1970		0.16 Hsiang-Tung, 1944			
Phalanger *Trichosurus vulpecula*	3	3.5				0.54 Martin *et al.*, 1970	Lower T Rees and Hore, 1970 Martin *et al.*, 1970	8 Rees and Hore, 1970 Martin *et al.*, 1970	6 Rees and Hore, 1970 Martin *et al.*, 1970

Table I (continued)

Animal	Digital dexterity	Body weight kg	Number of fibers per tract	Largest fiber diameter μm	Average fiber size $\times 10^5$ K	Area of tract mm^2	Penetration down spinal cord	Lamina of axon terminals	
								deepest	densest
Pygmy shrew *Microsorex hoyi*	2	0.003		1 VERHAART, 1970		0.02 VERHAART, 1970			
Quokka wallaby *Setonix brachyurus*	3	3.5				0.10 WATSON, 1971	T7 WATSON, 1971	6 WATSON, 1971	5 WATSON, 1971
Rabbit (domestic) *Oryctolagus cuniculus*	2	1.8	101,700 LASSEK and RASMUSSEN, 1940	4 VERHAART, 1970	0.27	0.28 LASSEK and RASMUSSEN, 1940	C4 VERHAART, 1970		
Raccoon *Procyon lotor*	3	12					S SIMPSON, 1912	10 PETRAS and LEHMAN, 1966	
Rat (unspecified)	3	0.4	137,000 BROWN, 1971	5 BISHOP *et al.*, 1971	0.10	0.14 BROWN, 1971	S GOODMAN *et al.*, 1966	8 GOODMAN *et al.*, 1966	5 BROWN, 1971
Rock hyrax *Procavia capensis*	2	17		9 VERHAART, 1967		0.72 VERHAART, 1970	S VERHAART, 1967		6 VERHAART, 1967

Table I (continued)

Animal	Digital dexterity	Body weight kg	Number of fibers per tract	Largest fiber diameter μm	Average fiber size $\times 10^5$ K	Area of tract mm²	Penetration down spinal cord	Lamina of axon terminals: deepest	Lamina of axon terminals: densest
Seal *Callorhinus ursinus*	1	52	748,000 Lassek and Karlsberg, 1956	25 Lassek and Karlsberg, 1956; Verhaart, 1970	0.27	2.08 Verhaart, 1970			
Sheep (domestic) *Ovis aries*	1	80	238,000 Lassek, 1942	5 Verhaart, 1970	0.59	1.42 Lassek, 1942	C1 King, 1911		6 King, 1911
Slow loris *Nycticebus coucang*	5	1.3		4.5 Verhaart, 1970		0.28 Verhaart, 1966	S Campbell, 1967	10 Campbell *et al.*, 1966	6 Campbell, *et al.*, 1966
Spiney anteater *Tachyglossus aculeata*	2	4.2				0.39 Verhaart, 1970	T16 Goldby, 1939		
Squirrel *Sciurrus hudsonius*	3	0.6				0.40 Simpson, 1914	S Simpson, 1914		

Table I (continued)

Animal	Digital dexterity	Body weight kg	Number of fibers per tract	Largest fiber diameter μm	Average fiber size $\times 10^5$ K	Area of tract mm^2	Penetration down spinal cord	Lamina of axon terminals: deepest	Lamina of axon terminals: densest
Tapir (unspecified) *Tapirus*	1	250		5 Verhaart, 1970		1.48 Verhaart, 1970			
Tarsier *Tarsius*	4	0.12				0.16 Tilney, 1927			
Tasmanian potoroo *Potorous apicalis*	2	1.35					T12 Martin *et al.*, 1972	7 Martin *et al.*, 1972	5 Martin *et al.*, 1972
Tree shrew *Tupaia glis*	3	0.16		2 Verhaart, 1966		0.12 Tigges and Shantha, 1969	Lower T Shriver and Noback, 1967	6 Jane *et al.*, 1969	5 Campbell, 1967
Whale (porpoise) *Phocaena phocoena*	1	60		5 Verhaart, 1970		1.27 Verhaart, 1970			
Woodchuck *Marmota monax*		5.3				0.37 Douglas and Barr, 1950	S Douglas and Barr, 1950		

Parameters Describing the Animal's Body

Since the size of an animal sets limits on the size of its neurological structures, it was important to determine the influence of body size, as measured by weight and length, on the value of the other parameters discussed in this report.

Body weight. The body weights assigned to each animal were taken from one of four sources: (1) the same report as the neuroanatomical data; (2) *Mammals of the World* [WALKER, 1968]; (3) *A Handbook of Living Primates* [NAPIER and NAPIER, 1967]; (4) TOWE [1973]. Whenever two of the secondary sources disagreed on Primates, the figures of NAPIER and NAPIER [1967] were selected. For domestic animals, the weights given by TOWE [1973] were used. In cases in which only a range of weights was given (including cases of marked sexual dimorphism) the midpoint of the range was used for statistical purposes.

Body length. As with weight, the body length for each animal was obtained either from the relevant neuroanatomical reports or from standard references [NAPIER and NAPIER, 1967; WALKER, 1968]. Since, as it turned out, this measure added nothing to the analysis beyond that contributed by body weight, no further mention of it seems warranted.

Parameters Describing the Pyramidal Tract

Tract size. The size of the pyramidal tract varies greatly among mammals. This variation has engendered considerable comment since the tract's discovery by PETIT in 1710. Recently, VERHAART [1970] has suggested that the size of the tract is a particularly important factor in determining its overall level of development.

As a measure of the size of the pyramidal tract, the area of a single pyramid at the level of the lower medulla has been used here. When the area of the tract at the lower medulla was not available, a reasonable estimate was made by one of several techniques. For example, if an author did not give the measurement but did provide a photomicrograph of the appropriate level and stated the magnification, we traced the border of the tract and then measured its area with a planimeter.

Nevertheless, several cases required special treatment: guinea pig, squirrel, and four bats. For guinea pig and squirrel an average of two measures had to be taken, the first from the upper medulla and the second from the upper cervical spinal cord. For two bats, *Harpyiocephalus* and *Pteropus,* the only measures available were from the lower pons and these were used despite being an obvious overestimate. For two other bats, *Desmodus* and *Rhinolophus,* no pyramidal tract has been reported even at the level of the upper pons. But since it was necessary for some potentially interesting comparisons to avoid absolute zero, we arbitrarily assigned a very small non-zero value: 0.01 mm^2. But because of the large number of mammals for which a reasonably accurate estimate of pyramidal tract size is available (48), these inaccuracies in the estimates for bats have very little statistical effect and no effect whatever on the main conclusions.

Number of fibers in the tract. The number of fibers in the pyramidal tract is known to vary widely among mammals but there has been disagreement as to the importance of this factor. LASSEK [1954], in his review, concluded that neither size nor skill of an animal was related to the number of fibers in its pyramidal tract. Apparently, it was this belief that prompted him to state that the function of the pyramidal tract is an enigma. TOWE [1973], however, has found a strong relation between the number of fibers in the pyramidal tract and both the size of an animal and the number of neurons in its cortex.

As estimates of the values for this controversial parameter, published fiber counts taken from the lower medulla were used. The difficulty in comparing fiber counts from different authors using different histological techniques has been well described by TOWE [1973]. The procedure used here for deriving a single estimate of the number of fibers per pyramidal tract is similar to TOWE's, and the results are virtually identical in all but the very few cases where more recent data have become available. The greatest discrepancy is for rat and is probably due to the greater number of small fibers revealed by the electron microscopic study of BROWN [1971]. In general, when more than one estimate was available, electron microscope studies were given preference over light microscope studies, studies using fiber stains were given preference over those using myelin stains, and more recent studies were used in preference to early ones, providing explicit citation of the earlier studies was made. When there were several comparable figures on the same species, the average was used.

Diameter of largest fibers. Including the diameter of the largest axons in the pyramidal tract in our analysis allowed exploration of the possibility suggested by electrophysiological reports that variations in digital dexterity are in direct correspondence with the existence of a large-fiber fast-conducting system [BERNHARD and BOHM, 1954; PRESTON *et al.*, 1967]. The values used for this measure were obtained from the original anatomical reports. As before, when several comparable figures were available, those from more recent papers were selected.

Average fiber size. The most refined description of the pyramidal tract would include the distribution of diameters of its constituent fibers – an obviously important potential factor in exploring its role in digital dexterity. Although few investigators have described the entire spectrum of fiber sizes in their subjects' pyramidal tracts, it is possible to derive a loose estimate of average fiber size by dividing the area of the tract by the number of fibers within it. As TOWE [1973] has pointed out, this calculation overestimates true average fiber size by about 2.2 times. Nevertheless, since the correlation techniques used here serve to remove this source of constant bias, it seemed worthwhile to examine the possibility that this feature might be in close correspondence with digital dexterity despite the inaccuracy in its estimation.

Parameters Describing the Locus of Termination of the Pyramidal Tract in the Spinal Cord

Descriptions of the terminations of the pyramidal tract within the spinal cord constitute an entirely different class of variations in form of the pyramidal tract – one which has considerable *a priori* potential as a factor underlying digital dexterity. Our ability to include these new dimensions in our analysis is the direct result of the recent renewal of activity in pyramidal tract anatomy fostered by the invention of the reduced silver technique [NAUTA and GYGAX, 1954] and its several modifications [FINK and HEIMER, 1967].

Depth of penetration down the spinal cord. This parameter encodes the most caudal segment of the spinal cord to which pyramidal tract axons have been traced and therefore is a measure of the relative length of the pyramidal tract. Some measurement error was introduced by the necessary inclusion of data obtained from studies using only normal material, and from experimental studies using stains for degenerating myelin, in addition to those using the Nauta techniques for preterminal degeneration or degenerating boutons. In cases where data from more than one type of study were available, preference was given to the more revealing techniques. That is, data based on bouton-degeneration were used if available. If not, then data from preterminal degeneration studies, myelin degeneration studies, or normal material were used in that order.

For statistical purposes, numerical rankings had to be assigned to the various levels of the spinal cord. These rankings were arbitrary but systematic. Depth of penetration was ranked from 1 to 9 according to the following rule: fibers ending at the upper cervical level were given the rank of 1; those ending at the lower cervical level ranked 2; upper thoracic ranked 3; middle thoracic ranked 4; lower thoracic ranked 5; upper lumbar ranked 6; lower lumbar ranked 7; sacral ranked 8 and coccygeal ranked 9.

Deepest and densest lamina of spinal gray receiving pyramidal tract termination. Many investigators have suggested that in those animals with a high degree of digital dexterity, the pyramidal tract is more likely to synapse on digital motoneurons [KUYPERS, 1964; BUXTON and GOODMAN, 1967]. Certainly, the synaptic sites of pyramidal tract axons have an obvious bearing on the type of influence they exert in the spinal cord. To investigate this potential relationship to digital dexterity, the lamina of pyramidal axon termination in the spinal cord gray was obtained and ranked using the system of REXED [1954]. The ventralmost, or *deepest,* lamina receiving pyramidal tract fibers and the lamina receiving the *densest* projections were recorded separately. For statistical purposes the laminae were ranked from dorsal to ventral with lamina I being ranked 1, lamina II ranked 2, and so forth through lamina VIII. REXED's lamina IX was divided into two ranks. A rank of 9 was used when there were but a few terminals reaching into lamina IX and these synapsing on dendrites. A rank of 10 was used only when there were dense terminals in lamina IX with some known to synapse directly on the soma of motor neurons.

Scale for Digital Dexterity

At present there is no commonly used index of digital dexterity that is applicable to all mammals. Therefore, we were forced to derive a suitable scale of our own (fig. 1). The dexterity scale in figure 1 is an extension of one invented by NAPIER [1961] and NAPIER and NAPIER [1967]. The scale ranks animals from 1 to 7 mostly on the basis of the anatomy of their hand. At one extreme, a rank of 1 indicates an animal with very limited ability to manipulate objects manually because of fusion of its phalanges (e.g. horses). At the other extreme, a rank of 7 indicates an animal (man) with a truly opposable thumb and capable of a 'precision grip' [NAPIER, 1960].

It should be noted that this anatomical rather than truly behavioral measure of dexterity probably underestimates the ability of some mammals. Perhaps a critical example of such underestimation is the case of the raccoon which has convergent but nonprehensile digits [NAPIER, 1961] and therefore has a dexterity rank of 3

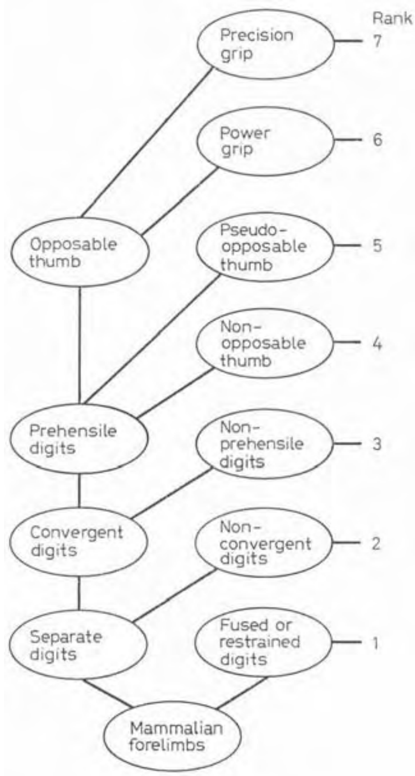

Fig. 1. Algorithm for ranking the digital dexterity of extant mammals – after NAPIER [1961] and NAPIER and NAPIER [1967].

for the purpose of this report. This rank indicates that the hand of the raccoon is in the same class as the hands of oppossums and squirrels. In opposition to this relatively lowly ranking on the basis of hand anatomy alone, it is not difficult to find agreement that behaviorally, the raccoon may be much more like monkeys (which have a rank of 5 or 6) in that they manipulate objects quickly and apparently dexterously. Nevertheless, we have adhered to the indicated ranking because cases like the raccoon tend reduce the correlations that are the substance of this report, since neuroanatomically the raccoon is also more like monkeys than like squirrels or oppossums. Therefore, our adherence to a strictly anatomical estimate of dexterity leads to more conservative conclusions than the alternative.

Statistical Adequacy of the Sample

The 69 species of mammals for which relevant pyramidal tract data are available can be considered, for statistical purposes, to be a sample (of size 69) of ex-

Table II. Population and sample distributions of extant mammals in genera per order

Order	True distribution of genera		Sample distribution of genera		Sample distribution – true distribution
	number	%	number	%	%
Monotremata	3	0.3	1	1.5	+1.2
Marsupalia	81	7.8	4	6.1	−1.7
Insectivora	66	6.3	5	7.6	+1.3
Dermoptera	1	0.1	0	0.0	−0.1
Chiroptera	180	17.3	6	9.1	−8.2
Primates	62	5.9	16	24.2	+18.3
Edentata	17	1.6	2	3.0	+1.4
Pholidota	1	0.1	1	1.5	+1.4
Lagomorpha	10	1.0	1	1.5	+0.5
Rodentia	369	35.4	10	15.2	−20.2
Cetacea	34	3.3	1	1.5	−1.8
Carnivora	102	9.8	6	9.1	−0.7
Pinnipedia	20	1.9	1	1.5	−0.4
Tubulidentata	1	0.1	1	1.5	+1.4
Proboscidea	2	0.2	1	1.5	+1.3
Hyracoidea	3	0.3	1	1.5	+1.2
Sirenia	3	0.3	0	0.0	−0.3
Perissodactyla	6	0.6	1	1.5	+0.9
Artiodactyla	82	7.9	8	12.1	+4.2
Total	1043	100.2	66	99.9	

tant mammalian species; or since three genera are each represented here by two species, it can be considered to be a sample (of size 66) of extant mammalian genera. This latter description allows a determination of the adequacy of the sample for deriving conclusions about the entire population of mammalian genera because the natural distribution of mammalian genera is known [WALKER, 1968]. Comparing the present sample with the natural population, it is found that the sample deviates from the natural distribution by less than 5% for every order except Primates, Rodentia, and Chiroptera (table II). Primates are nearly four times as numerous in the sample as in the natural population, while Rodentia and Chiroptera are each underrepresented by about half. Although this emphasis on Primates is useful for the purposes of this report, for strictly statistical purposes, it is a simple sampling bias. Therefore, the animals used here as a sample of the class Mammalia should not be considered to be subject to merely random sampling error, and inferences from it to the entire population of mammals must be weakened accordingly.

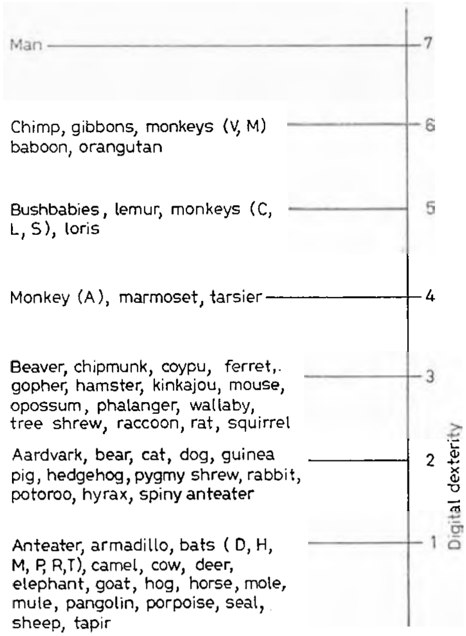

Fig. 2. Distribution of digital dexterity among 69 mammals as scaled by the rule shown in fig. 1.

Results and Discussion

The raw data on which the statistical computations have been based are shown in table I, the initial correlation coefficients are shown in table III, partial correlations in table IV, and the distribution of digital dexterity in figure 2; and the distributions of digital dexterity and seven morphological parameters are shown in figures 2–9.

Relation of Digital Dexterity to the Size of the Pyramidal Tract and its Constituent Fibers

Size of the pyramidal tract. The intuitive idea that the pyramidal tract might be better developed in dextrous animals would seem to require little argument, although this was not true 20 years ago [Lassek, 1954; Towe, 1973]. The distribution of this parameter for the 54 mammals for which data are available is shown in figure 3. Man has the pyramidal tract with the largest area (11.43 mm^2) while mole and some bats have the smallest (0.01 mm^2 at most).

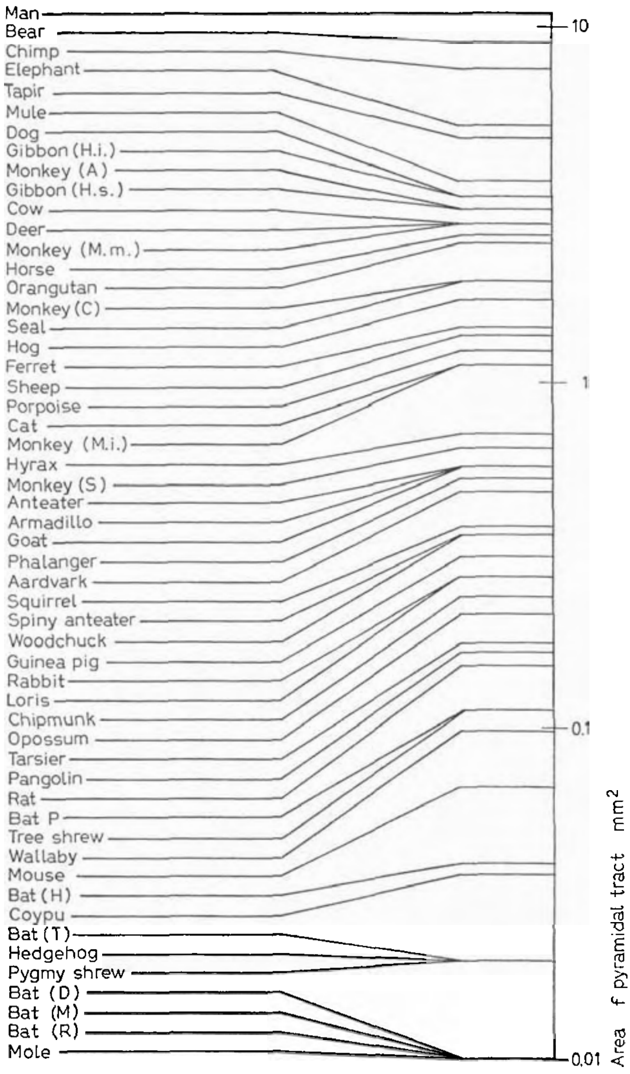

Fig. 3. Distribution of area of pyramidal tract in 54 mammals. Note nondextrous animals (elephant, tapir, mule, etc.) near top of list along with man and chimpanzee.

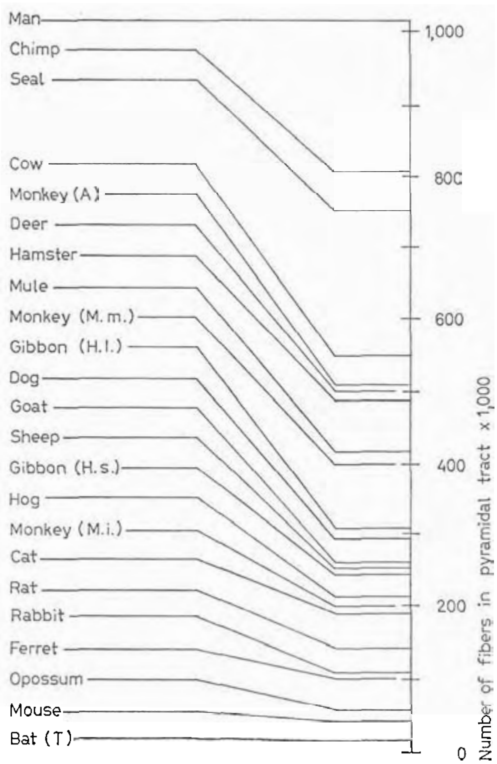

Fig. 4. Distribution of the number of fibers in the pyramidal tract in 23 mammals. Note nondextrous animals (seal, cow) above monkeys.

Since the extremes of this scale match the extremes of the digital dexterity scale shown in figure 2, the idea that the two are related is promptly supported. However, the correlation between the two scales proves to be a disappointingly low value of only 0.34 (table III) which, though statistically reliable, would account for less than 12% of the variation in dexterity. The reason for this low correlation can be seen in figure 3, where bear intercedes between man and chimpanzee; elephant, mule, and dog are above the monkeys; and four ungulates are above one Rhesus monkey.

Close scrutiny of figure 3 gives the impression that body size, not digital dexterity, is the closest correlate of pyramidal tract size with larger animals near the top of the distribution and smaller animals near the bot-

Table III. Correlation coefficients of morphological parameters with

		Digital dexterity	Body parameters	
			log body weight	log body length
Body parameters	log body weight	–0.09 (69)		
	log body length	–0.08 (69)	0.94** (69)	
Tract parameters	log number of fibers	0.26 (23)	0.62** (23)	0.57** (23)
	largest fiber diameter	0.26 (31)	0.34 (31)	0.38* (31)
	log average fiber size (area number)	0.42 (21)	0.48* (21)	0.47* (21)
	log area	0.34* (52)	0.81** (53)	0.77** (53)
Termination parameters	penetration down spinal cord	0.71** (44)	–0.28 (45)	–0.23 (45)
	deepest lamina	0.66** (25)	0.25 (25)	0.24 (25)
	densest lamia	0.35 (20)	0.49* (20)	0.22 (20)

Numbers in parentheses designate number of animals contributing to correlation
* = $p < 0.05$; ** = $p < 0.01$.

tom. Calculation of the correlation coefficient of body weight and pyramidal tract size confirms this impression – it is 0.81 (table III), a very high and very reliable value indicating that body weight alone accounts for more than 65% of the variation in pyramidal tract size. Therefore, the area of the pyramidal tract depends more on an animal's body size than on its dexterity; large animals tend to have large pyramidal tracts whether or not they are dexterous. This conclusion is in agreement with TOWE [1973].

That this close correspondence between body size and tract size effectively masks the relationship between digital dexterity and tract size can be demonstrated by removing body size as a factor in the correlation between tract area and dexterity. This calculation results in an increase in the correlation to 0.71 (table IV) which means that there does exist a relationship between tract size and dexterity, although it is all but smothered by the relationship between tract size and body size. Therefore, it can be concluded that dextrous mammals tend to have large pyramidal tracts. However, because a large pyramidal tract accompanies either large bodies or high digital dexterity, a large pyramidal tract is not necessarily a sign of high dexterity.

digital dexterity and intercorrelations among morphological parameters

Tract parameters				Termination parameters	
log number of fibers	largest fiber diameter	log average fiber size	log area	penetration down cord	deepest lamina
0.75** (14)					
0.50* (21)	0.15 (13)				
0.93** (21)	0.54** (28)	0.78** (21)			
0.37 (15)	0.49* (21)	0.33 (14)	0.22 (32)		
0.54 (9)	0.52 (10)	0.58 (9)	0.68** (17)	0.62** (24)	
0.82* (8)	0.37 (10)	0.67 (8)	0.77** (16)	0.59** (19)	0.78** (18)

coefficient.

Number of fibers in pyramidal tract. Since the overall size of the pyramidal tract is not a very good indicator of digital dexterity, perhaps dexterity depends on the nature of its constituent fibers. Perhaps the number of fibers in the tract is the key: do more dextrous animals have more fibers? The distribution of the number of fibers in the pyramidal tract for the 23 mammals on which fiber counts are available, is shown in figure 4. The pyramidal tract of the bat *(Tasarida cynocephala)* has the least number of fibers (8000) while man has the most (1,100,000).

Since man and chimpanzee are at the top of the distribution as they are for digital dexterity and bat is at the bottom of both distributions, it is tempting to presume that the number of pyramidal tract fibers is indeed related to digital dexterity. However, the correlation between the two parameters is low once more, only 0.26 (table III) which, even if it were statistically reliable, would account for less than 7% of the variation in digital dexterity.

The reason for this low correlation can be seen in figure 4. The ungulates, cow, deer, mule, goat, sheep, and hog, have widely divergent numbers of fibers and yet all are hoofed and rank lowest in digital dexterity. Similarly, the New World owl monkey without an opposable thumb

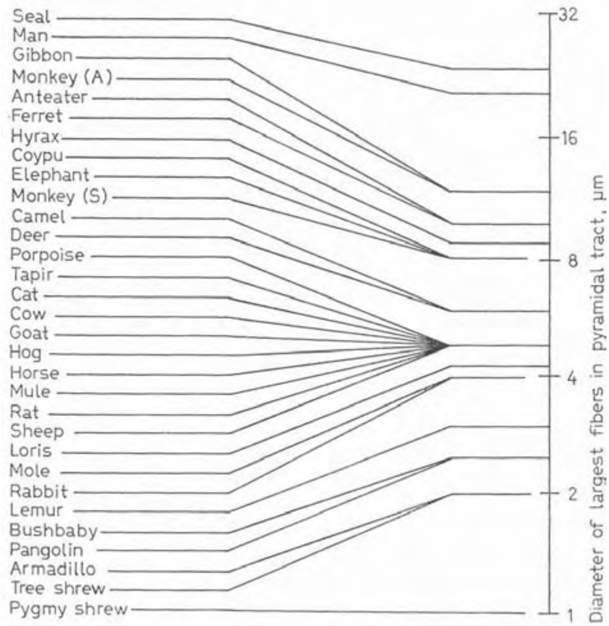

Fig. 5. Distribution of the diameters of the largest fibers in the pyramidal tract in 31 mammals. Note nondextrous animals near top of scale.

ranks well ahead of the Old World monkeys *(M. mulatta* and *M. irus)* and gibbons, even though the latter have a higher ranking for digital dexterity. Obviously, a direct relation between the number of pyramidal tract fibers and digital dexterity is not indicated.

Before abandoning the possibility that the number of pyramidal tract fibers might be related to digital dexterity, however, it can be noticed that the distribution in figure 4 shows, again, a parallel to body size – with smaller animals (bat, mouse, etc.) near the bottom and larger animals (man, chimp, seal) near the top. Indeed, the number of pyramidal tract fibers, much like the size of the tract itself, is much more closely and reliably correlated with body weight (0.62, table III) than with dexterity.

This observation suggests that there may exist at least two sources of selective pressure on the number of fibers in the pyramidal tract, body size and digital dexterity, and that the former may mask the latter once more. To examine this possibility further, the correlation of the number of fibers with digital dexterity was calculated holding body weight con-

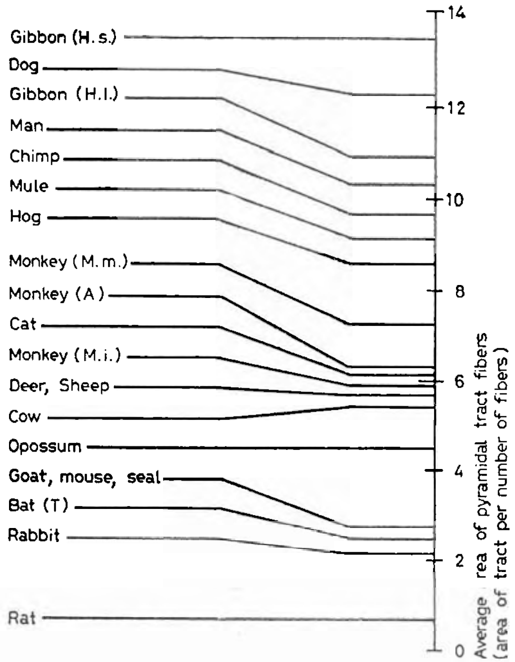

Fig. 6. Distribution of derived estimate of average size of fibers in pyramidal tract in 21 mammals. Note nondextrous animals near top of scale. Indicated value probably overestimates average fiber size by about 2.2 times [Towe, 1973].

stant. This partial correlation proves to be 0.40 (table IV) – still too small to be statistically reliable with 23 cases, but noticeably higher than the previous correlation. This result indicates that there may be a relationship between digital dexterity and the number of fibers in the pyramidal tract; but if there is, it is a very subtle one, accounting for only about 16% of the variation in digital dexterity among mammals and requiring several more supporting cases and no contradictory ones before gaining an acceptable level of reliability.

Spectrum of fiber sizes in the pyramidal tract. If it is not the size of the pyramidal tract nor the number of fibers within it that is most directly related to digital dexterity, is it perhaps fiber size? Do dextrous mammals have generally larger fibers or do they have an extra component in their pyramidal tract – perhaps a system of large diameter, fast conducting fibers?

Table IV. Partial correlations of digital dexterity with pyramidal tract parameters

Factors statistically removed	Digital dexterity by						
	log number of fibers	largest fiber	log average fiber size	log area	penetration down cord	deepest lamina	densest lamina
1 log body weight	0.40	0.31	0.53*	0.71**	0.72**	0.71**	0.45*
2 log number of fibers		0.10	0.35	0.28*	0.68**	0.64**	0.25
3 Largest fiber diameter	0.10		0.40	0.25	0.69**	0.64**	0.28
4 log average fiber size	0.06	0.22		0.02	0.67**	0.56**	0.10
5 log area	−0.16	0.10	0.26		0.69**	0.62**	0.15
6 Penetration down cord	0.00	−0.14	0.28	0.27*		0.40*	−0.12
7 Deepest lamina	−0.15	−0.13	0.06	−0.20	0.51**		−0.35
8 Densest lamina	−0.05	0.15	0.27	0.12	0.67**	0.66**	

The distribution of the diameters of the largest fibers in the pyramidal tract for 31 mammals is shown in figure 5. The pygmy shrew has the smallest large fibers (about 1 μm in diameter) and seal has the largest (about 25 μm).

Although man is not at the top of this distribution, he might be considered sufficiently close to the top to pursue a potential relationship with dexterity nevertheless. However, the correlation between the size of the largest fibers and digital dexterity is only 0.26 (table III), again disappointingly small and statistically unreliable. Even if the correlation were reliable, this relationship would account for less than 7% of the variation in dexterity. Furthermore, when body weight (which is loosely related to the presence of large fibers, too) is removed as a factor mathematically, the partial correlation between size of the largest fibers and digital dexterity rises only to 0.31 (table IV) which, if it were also not unreliable, would account for only 9% of the variation in dexterity.

Therefore, we are forced to conclude that the presence of large fibers in the pyramidal tract, like the total number of fibers and the overall size of the pyramidal tract, does not go very far in accounting for digital dexterity. Clearly, the presence of a large-fiber component in the pyramidal tract is neither an indication of digital dexterity nor vice versa.

Average fiber size. Finally, the distribution of average fiber size in the pyramidal tract of 21 animals is shown in figure 6. Man is far from the top of the distribution, and animals such as dog, mule, and hog have larger fibers than macaque monkeys. As a result, the correlation be-

Pyramidal Tract

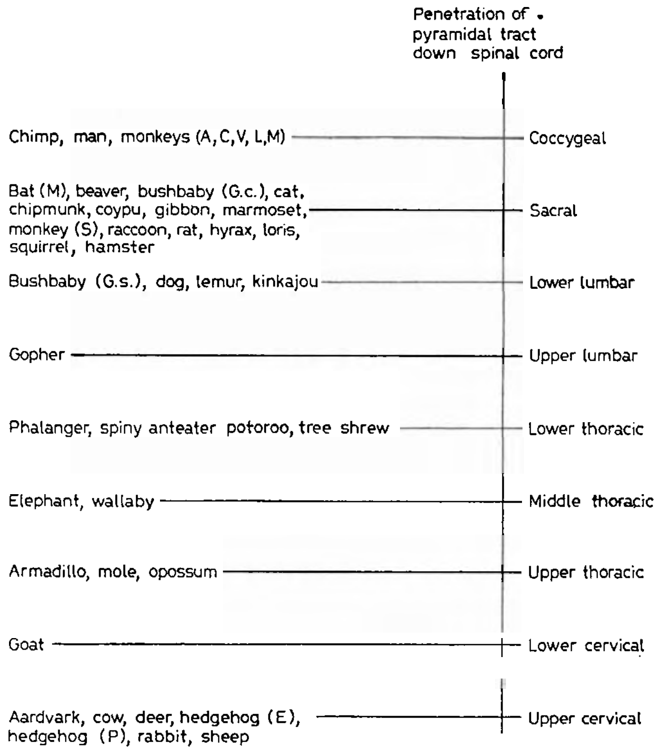

Fig. 7. Distribution of penetration or extension of pyramidal tract down spinal cord in 44 mammals. Note concentration of dextrous animals near top of scale. This parameter shows the closest correspondence with digital dexterity of any included in this report.

tween average fiber size and digital dexterity is 0.42 (table III) and statistically unreliable. Upon closer inspection of figure 6, it can be seen that the larger animals tend to be nearer the top than smaller animals. Indeed, when body weight is held constant, the correlation increases to 0.53 (table IV), a value which is reliable. Thus, as was the case with area of the tract (on which average fiber size partly depends), a loose relation between average fiber size and dexterity is indicated even though it is concealed by their mutual accompaniment of large bodies.

However, because average fiber size was calculated from tract area and number of fibers (average fiber size = area of tract/number of fi-

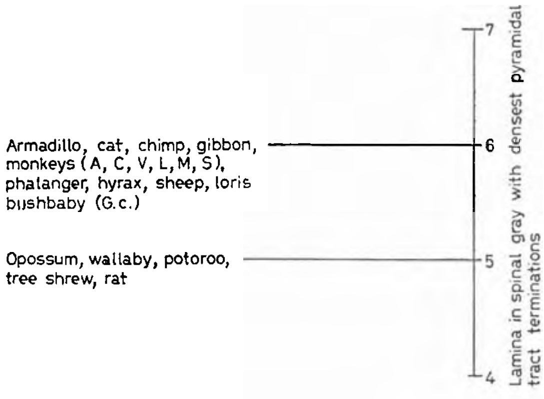

Fig. 8. Distribution of lamina of spinal cord gray matter receiving most pyramidal tract terminations in 20 mammals.

bers) and because we have already seen that tract area is related to dexterity while number of fibers is not, this derived relation between dexterity and average fiber size probably adds no new dimension to the analysis. This idea is upheld by the decrease in the correlation of dexterity with average fiber size when tract area is held constant and vice versa. This result means that either the area of the tract or average fiber size as derived here may be a modest factor in digital dexterity, but both seem to be estimating essentially the same feature. Because values for the area of the tract were determined empirically and because its relationship with dexterity is stronger, the relation between dexterity and average fiber size would not seem to warrant further discussion at this time.

In summary, digital dexterity is related to pyramidal tract size and modestly related to average fiber size even though each of these two parameters is more closely related to an animal's body size. In contrast, no important or direct relationship seems to exist between digital dexterity and either the total number of fibers in the tract or the presence of a large-fiber component.

Relation of Digital Dexterity to the Sites of Termination of the Pyramidal Tract

In the previous section, the relationship of digital dexterity to the size of the pyramidal tract and its constituent fibers has been examined and

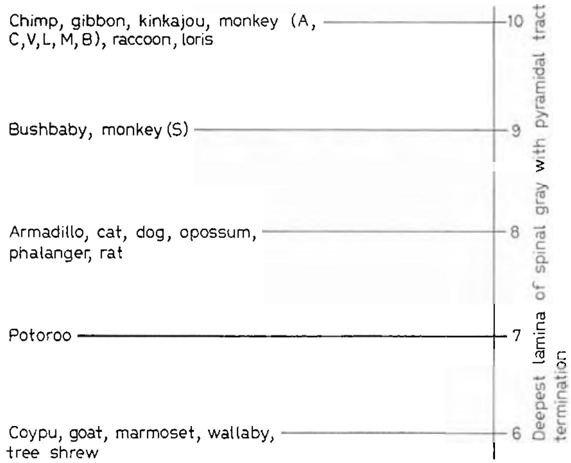

Fig. 9. Distribution of deepest or ventralmost lamina of spinal cord gray matter receiving pyramidal tract terminations in 25 mammals. Note absence of non-dextrous animals from top of scale. Rank 10 assigned to animals known to have direct pyramidal projections onto motoneuron somata.

only two relationships were found. However, neither of these relationships seems sufficiently strong to merit special attention. Certainly, they are not morphological requisites for dexterity. Therefore, the question turns to an entirely different class of morphological variations: if it is not the form or the fibers in the tract that are related to digital dexterity, could it be their manner of termination in the spinal cord?

In exploring this possibility, three different sets of data describing the variation in spinal cord terminations were collected: (1) the depth of penetration of the pyramidal tract down the spinal cord; (2) the lamina of the spinal gray receiving the most pyramidal tract terminations, and (3) the deepest lamina in the spinal gray receiving a significant number of pyramidal tract terminations. In this section, the relation of digital dexterity to each of these parameters is discussed in turn.

Depth of penetration of pyramidal tract down spinal cord. The basis for the idea that digital dexterity might be related to the depth of penetration of the pyramidal tract can readily be seen in figure 7. In man, chimpanzee, and most monkeys, the pyramidal tract extends the full length of the spinal cord, while in cow, deer, sheep, hedgehog and aardvark the tract does not extend beyond upper cervical levels. Among the

44 mammals on which data are available, each of the primates is in the top half of the list while each of the ungulates is in the bottom half.

This apparent correspondence between the extent of the tract and digital dexterity is upheld by the correlation coefficient between the two rankings which is 0.71 (table III) – a notably high and statistically reliable value which accounts for about 50% of the variation in digital dexterity. The reason why this value is so high can be seen in figure 7 where the bottom of the list consists entirely of nondextrous animals.

However, the reason why this value is not still higher can also be seen in the figure where several nondextrous mammals (bat, beaver, hyrax, etc.) appear high in the list. This result means that dextrous animals have long pyramidal tracts, but a long pyramidal tract does not guarantee dexterity. Therefore, it can be concluded that selective pressure on digital dexterity may be one major source of selection pressure on the depth of penetration of the pyramidal tract down the spinal cord – perhaps even the greatest one – but it is not the only one. The existence of several nondextrous mammals with long pyramidal tracts indicates that there is at least one source of selective pressure on the extent of the pyramidal tract other than digital dexterity. What this other source might be, the present data do not reveal. Nevertheless, the relationship between the relative extent of the pyramidal tract and digital dexterity is by far the closest one discussed so far.

Lamina of spinal cord gray receiving the most pyramidal tract terminations. Since the advent of experimental neuroanatomical and physiological recording techniques, investigators have discovered that most of the fibers in the pyramidal tract do *not* terminate directly on motoneurons in the ventral horn in any animal. Instead, the terminations extend from REXED's lamina II to lamina VI in all mammals yet examined, from lamina II to lamina VII or VIII in several cases, and from lamina II to lamina IX in only a few cases. Of course, it is this persistence of pyramidal tract terminations in the upper, sensory laminae of the spinal gray that gives support to HUGHLINGS JACKSON's idea [TAYLOR, 1931] that the pyramidal tract is primarily a part of the somatosensory system. But for present purposes, it is the variation of the ventral terminations that is of special interest.

The distribution of the lamina receiving the most projections from the pyramidal tract in the 20 mammals for which data are available is shown in figure 8. It can be seen that in each of the 20 animals the tract projections are heaviest or most dense in either lamina V or lamina VI.

No mammal yet studied concentrates its pyramidal tract projections in a lamina other than one of these two. Although this consistency is interesting in its own right because of its implications for pyramidal tract-spinal cord physiology, for the present purpose it is a very narrow and unrefined range of variation – apparently too unrefined to account for the range of digital dexterity.

Even though each of the dextrous animals (monkeys and chimpanzee) have their densest projections to lamina VI, it can be seen that several nondextrous animals (i.e. armadillo, hyrax, sheep) do, too. And, of course, the absence of data on man raises the suspicion that the distribution may be truncated. This general impression is supported by the correlation between this and the dexterity scale which is both low and unreliable (0.35, table III).

Therefore, given the present amount of data, the lamina of spinal cord gray receiving the most or densest pyramidal tract terminations is not a useful indicator of digital dexterity.

Deepest lamina of spinal gray receiving pyramidal tract terminations. Turning now to the final parameter describing the locus of pyramidal tract terminations, an entirely different result obtains. Figure 9 shows the distribution of the deepest or ventralmost lamina of the spinal gray which receive pyramidal tract terminations. Of the 25 mammals for which data is available (man is missing once more), the most dextrous have the most ventral terminations. This is the relationship suggested by Kuypers [1964] on the basis of his comparison of rat, cat, and monkey. Indeed, using a rank of 10 to signify direct projections onto the somata of motoneurons, it can be seen that each of the Primates is ranked either 9 or 10 and the only nonprimates with that ranking are two carnivores which are the most dextrous of their order (kinkajou and raccoon). Thus, the correspondence between the deepest lamina receiving pyramidal tract terminals and digital dexterity is not confined to Primates but also serves to indicate the dexterity of Carnivora. This impression is supported by the correlation coefficient between this ranking and dexterity, which at 0.66 (table III) is both high and reliable.

Therefore, it can be concluded that the deepest lamina of pyramidal tract termination is a good indicator of digital dexterity in all mammals and an especially good indicator in the extreme cases. Indeed, unless man proves to have no pyramidal tract terminations below lamina VIII, it is difficult to see how this close correspondence could be seriously weakened.

Prediction of Digital Dexterity from Pyramidal Tract Morphology

In the previous sections it has been shown that two features of pyramidal tract morphology bear a strong relationship with digital dexterity while two other intercorrelated ones are modestly related. In order to pursue these relationships one step further, a final statistical procedure was carried out in an attempt to relate an animal's dexterity ranking to a combination of these parameters. This multiple regression procedure derives the best equation between dexterity, on one hand, and a linear combination of the morphological parameters on the other. Four parameters were included in the analysis: 'deepest lamina of spinal gray receiving pyramidal tract terminals', 'penetration of tract down the spinal cord', 'area of tract', and 'body weight'. ('Body weight' was included simply because it was available for each animal; while 'average fiber size' was excluded due to its redundancy with tract size.)

There were 17 animals of the original 69 for which data were available on each of the four features: armadillo, cat, chimpanzee, coypu, dog, gibbon, goat, owl monkey, cebus monkey, macaque monkey, squirrel monkey, opossum, phalanger, wallaby, rat, slow loris, and tree shrew. Consequently, the multiple regression equation that was derived is based on a sample far from the randomness required for its generalization to all mammals. Nevertheless, the outcome provides a reasonable first approximation to the particular combination of morphological features that best predict digital dexterity.

When the digital dexterity ranking and the four morphological parameters are each transformed to standard normal variables, the best function is:

$$\text{Dexterity} = (0.65)D + (0.49)P + (-0.30)A + (0.16)W$$

where 'D' is the deepest lamina receiving pyramidal tract terminals, 'P' is the relative extent of the tract down the spinal cord, 'A' is the log area of the tract, and 'W' is the body weight of the animal.

The analysis shows that this equation yields a dexterity prediction which has a correlation coefficient with true digital dexterity of 0.815 which would account for more than 66% of its variation in the sample. Although this correlation is notably high and reliable, it is still too low to be useful for individual cases. As data on a wider variety of mammals become available, it can be expected that this compound relationship can be made more accurate and potentially more useful.

However, mere acquisition of more data on the same parameters cannot be expected to account for all the variation in dexterity. For example, at least one entire class of parameters (describing the origin of the tract) is missing from the present analysis. Until these dimensions and still others yet to be discovered are included, an account of the factors underlying digital dexterity will remain incomplete. At this time, therefore, the compound equation given above serves only to illustrate the main point once more: it is the mode of termination of the pyramidal tract, rather than the size of the tract or its constituent fibers, that is most closely related to digital dexterity.

General Discussion

In the previous section we have seen that among the eight morphological dimensions of pyramidal tract variation included in the analysis, only two (penetration down spinal cord and deepest lamina receiving pyramidal terminations) show a reasonably close and reliable correspondence with digital dexterity while two others (the overall size of the pyramidal tract and the average size of its fibers) show a reliable but much more modest correspondence. The first three of these relationships are shown in figures 10–12.

The chief outcome of the entire analysis is that the two parameters showing the closest correspondence with digital dexterity both focus on the terminations of the pyramidal tract rather than on the size of the tract or on the size, spectrum, or number of fibers within it. Therefore, it is reasonable to conclude that it is not so much the nature of the tract or its constituent fibers that is important for digital dexterity, but, instead, where the fibers terminate. If the tract extends to the lowest segments of the spinal cord and if the fibers terminate directly on spinal motoneurons, then the animal is very likely to be dextrous regardless of the size of the tract and regardless of the size and number of its constituent fibers.

It is easy to see how the directness of termination of pyramidal tract fibers on motoneurons might be related to digital dexterity and particularly to discrete and independent movements of the individual digits. Furthermore, it is easy to see how the overall size of the pyramidal tract or the average size of its constituent fibers might contribute to digital dexterity. However, it is not so easy to see the nature of the close relationship between dexterity and the extent to which the pyramidal tract penetrates the spinal cord. One might think that once the pyramidal

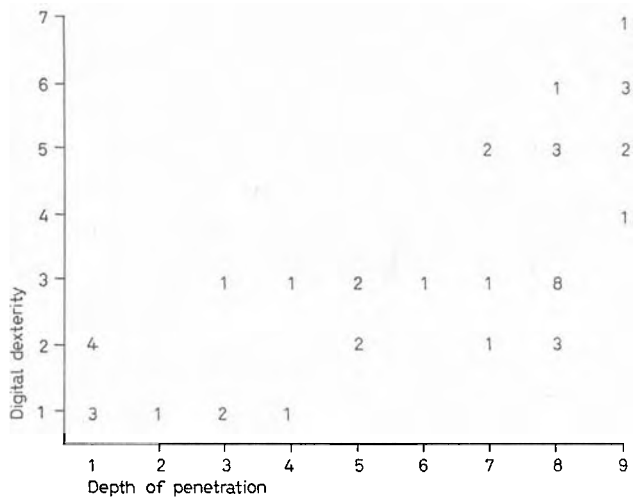

Fig. 10. Relationship of digital dexterity to depth of penetration of pyramidal tract down spinal cord. Numerals indicate number of different species at each point. Correlation coefficient is 0.71 (table III). When animal's body size is held constant, correlation is 0.72 (table IV).

tract extended beyond the cervical segments, further penetration would have little effect on manual dexterity. However, there are several plausible explanations for this strong relationship, three of which merit discussion.

To begin with, the relationship of the extent of the pyramidal tract to dexterity might be due to both their relationship to the lamina of termination, that is, one can envisage a progressive penetration of the cord that would bring with it an increase in the number of fibers terminating in lamina IX at the higher levels. If such were the case, the two rankings, penetration down the cord and deepest lamina with terminations, should themselves be correlated, and they are (0.62, table III).

The existence of this interrelationship between the two parameters that are, themselves, in closest correspondence with digital dexterity suggests that the two dimensions they describe might be collapsible into one, with an obvious gain in parsimony. To explore this possibility, we calculated the partial correlation of digital dexterity with each of the parameters while the other was held constant. These partial correlations turned out be 0.40 between dexterity and penetration with deepest

Pyramidal Tract

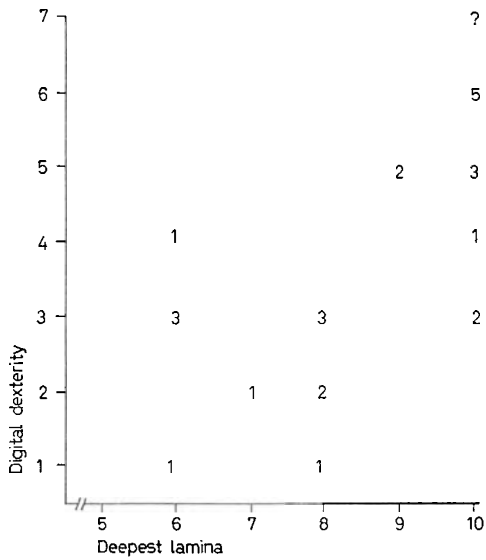

Fig. 11. Relationship between digital dexterity and the deepest or ventralmost lamina in spinal cord gray with pyramidal tract terminations. Numerals indicate number of different species at each point. Query indicates unverified point for man. Correlation coefficient (without man) is 0.66 (table III). When animal's body size is held constant, correlation is 0.71 (table IV).

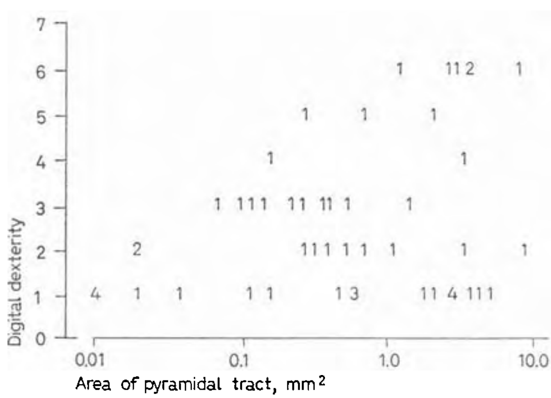

Fig. 12. Relationship of digital dexterity with area of pyramidal tract. Numerals indicate number of species at same or nearby points. Correlation coefficient is 0.34 (table III). When animal's body size is held constant, correlation is 0.71 (table IV).

lamina held constant; and 0.51 between dexterity and deepest lamina with penetration held constant (table IV). These values mean that despite the covariation of the two parameters, they each have an independent relationship with dexterity beyond that of the other. It follows that the two parameters are not collapsible into one and the possible contribution of each must be considered separately. Therefore, the question of how the extent of the pyramidal tract beyond the cervical levels might contribute to digital dexterity remains.

Perhaps the most obvious possibility is that the extent of the tract is not directly related to dexterity in a structure-function sense but is, instead, merely a correlate of some other feature that is related. For example, a pyramidal tract reaching the most caudal parts of the cord may be related more to dextrous hindfeet than forefeet. Since selective pressure for dextrous forefeet is probably not independent from pressure for dextrous hindfeet, the behavioral and morphological features would be correlated in a nonfunctional way.

On the other hand, it is possible that the two features are directly related – that digital dexterity requires cortical control over the entire spinal cord, not just over the cervical segments. For example, it is not unreasonable to suppose that the cortex must compete with intrinsic fiber systems of the spinal cord for control of cervical motoneurons and that one of the ancillary functions of the pyramidal tract in digital dexterity is to suppress these competing influences. If this were the case, one might expect that section of the corticospinal tract below the cervical level would result in a deficit in manual dexterity despite the preservation of direct pyramidal connections to cervical motoneurons. If this reasoning is correct, such a section would release the lower levels of the spinal cord from cortical suppression and allow abnormal interference with pyramidal control at cervical levels.

Whether or not this indirect role of the pyramidal tract in digital dexterity proves to exist, that high dexterity apparently occurs only in animals whose pyramidal tract extends beyond lumbar levels and synapses nearby or directly on motoneurons, are facts yet uncontradicted.

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